

HAT-P-37b

R-2238

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_260411 · created 2026-08-15T01:53:31+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461141.894 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.051
Transit depth (Rp/Rs)^2	2.91 +/- 1.72 [%]
Orbital Inclination (inc)	89.57 +/- 2.19
Airmass coefficient 1 (a1)	0.993 +/- 0.021
Airmass coefficient 2 (a2)	0.008 +/- 0.062
Scatter in the residuals of the lightcurve fit is	2.15 %
Best Comparison Star	#2 - [314, 394]
Optimal Method	PSF photometry
Transit Duration (day)	0.058 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.171 ± 0.051 vs 0.1394 (deviation 0.62σ)
Duration fit vs published	0.058 d vs 0.0983 d (deviation 3.36σ)
Mid-time O-C	-29.9 min
Depth SNR	3.4
Residual scatter	2.15 %

Light curve

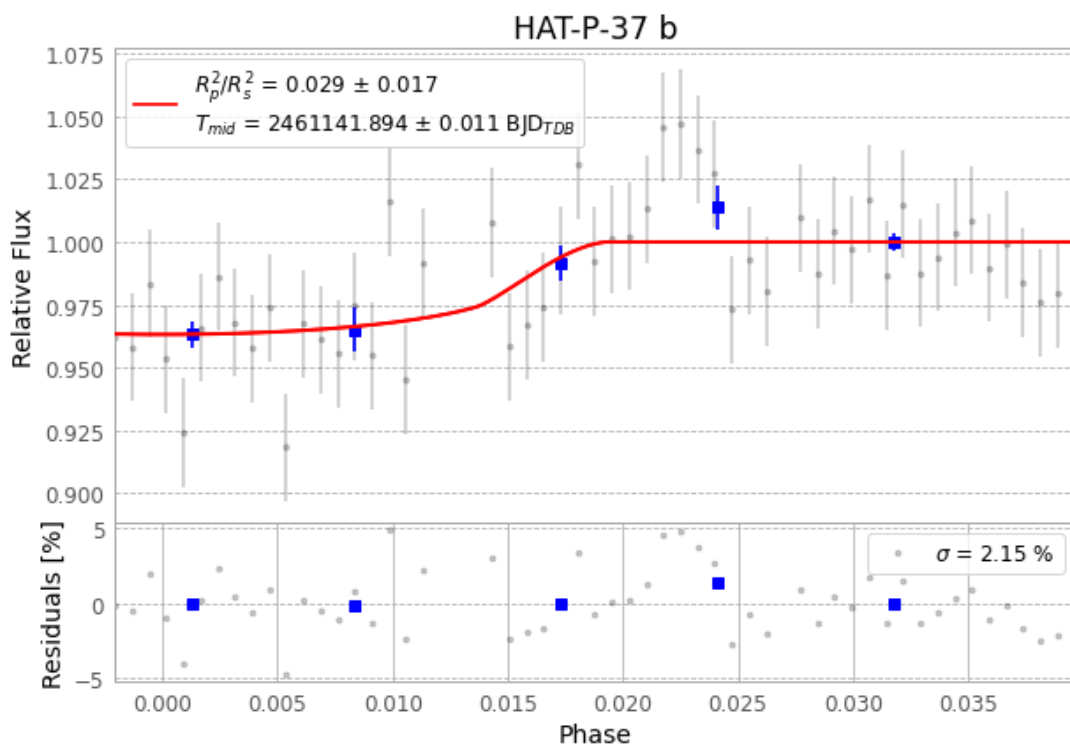


Figure 1 — FinalLightCurve_HAT-P-37 b_2026-04-11.png (SHA-256 5f88ae95564dfdf7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [293, 262], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 35b9d5d349e400ab...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

55 FITS frames (36 MB), HATP-37260411091816.FITS → HATP-37260411120615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-260411.log (c8594d18bd02...), FinalLightCurve_HAT-P-37 b_2026-04-11.png (5f88ae95564d...), FinalLightCurve_HAT-P-37 b_2026-04-11.csv (b3b9d68cae43...)

Compute resources

Wall time 177.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 01:53Z.